

data structure that you will use, which can store past queries and automatically provide suggestions? Will you store the complete query as is, or will you break it into its constituent words? Given that scalability and correctness is key for this problem, can you explain how the choice of your data structure will address both?

6. You are given a set of documents and you are asked to cluster them based on document level similarity. You are familiar with different clustering algorithms such as 'kmeans'; so you are confident that you can do the clustering. However, the interviewer now adds a second part to the problem, where he wants you to label the clusters. How would you generate labelled clusters? What are the means of generating meaningful labels for the clusters? Can the clusters be labelled post the clustering process or prior to it, or during the cluster process itself? Describe your choice of the solution and why you made that choice.
7. You have been asked to write code to reverse a singly linked list. You wrote a non-recursive version that can do this task. However, the interviewer contends that your solution is not good and a recursive solution will be better than a non-recursive solution in terms of time and space complexity. Do you agree with the interviewer? If yes, explain why? If no, can you explain the time and space complexity of non-recursive and recursive versions for reversing a linked list?
8. You are given an array of N records containing employee information, including the employee date of joining, name, address and salary. You are told to sort the array in ascending order in the order of employee date of joining. It is possible that multiple employee records may have the same joining date. In that case, the original order of these records should be retained in the final sorted array. (a) You are told that there is no restriction on the additional space that your algorithm can use. What is your solution? (b) You are told that you can only use constant additional storage space. What is your solution in this case?
9. You are given two arrays. You are asked to find the common elements between the two arrays. (a) Assume that the two arrays only contain integer elements. (b) Assume that the two arrays contain floating point values. Will your solution for (a) work for (b) as well?
10. Given an arbitrarily long character string, you are asked to find out the most frequently occurring character and the least frequently occurring character. Write a C function to solve this problem, given a character string of size N. Now, instead of a fixed size character string, you are given a stream of characters that are coming dynamically as input. At any point in time, you should be able to print the most frequent and least frequent characters in the stream (note that if two or more characters have the same highest frequency, you can choose to print any of them as the most frequent characters). What is the time and space complexity of your algorithm?
11. You have been given the problem of sentiment analysis, where given a sentence, you need to classify it as positive, negative or neutral sentiment bearing. You have been given large quantities of labelled data of example sentences, with their target sentiment polarity annotated. Given that there can be words in the test input sentences that are not present in the training set examples, how will your approach handle the out of vocabulary words in the test sentences? If you decide just to ignore the out of vocabulary words, what is the impact on the test set classification accuracy? Can you devise a better approach to handling this issue?
12. You are given a search query and you can use any of your favourite Web search engines to find the query results. Web search engines use their own ranking algorithm to rank the query search results. However, you are asked to apply a filter on top of the search engine query results such that the results are ordered temporally. For example, consider a query like: 'Joe Biden US Presidential election'. Given a set of search engine results for this query which will contain news articles, events, etc, you need to sort them temporally. For instance, a query result link on the news story from July 2019 describing Biden's intention to run should be ranked before the news story featuring his 2020 New Year wishes to the American electorate. Essentially, your task is to organise the search engine query results on a timeline. What are the challenges in organising the Web search engine query results temporally?
13. Consider that you are running your application on a desktop which has 16GB physical memory and is installed with the Linux OS. You want to determine the maximum memory footprint of your application. Explain how you will find out how much physical memory is being used by your application.
14. You are given an array of N integers. You are asked to find the largest element that appears an even number of times in the array. What is the time complexity of your algorithm? Can you do this without sorting the entire array?
15. In machine learning, what is the problem of generalisation? How is the learning model adapted to reduce the generalisation error as opposed to reducing the training or validation error? If a classifier has high cross-validation accuracy, does it follow that its test accuracy will also be high? If yes, explain why. If not, explain the reason why you think it may not be so.

Feel free to reach out to me over LinkedIn/email if you need any help in your coding interview preparation. If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Wishing all our readers happy coding until next month! Stay healthy and stay safe. **END** 🐧

By: Sandya Mannarswamy

The author is an expert in natural language processing and is currently working as an independent researcher. Her interests include natural language processing, machine learning and AI.